

UCC404: FUNDAMENTALS OF DATA SCIENCE

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Course Objective: To elaborate the basics of data science and provide a foundation for understanding the challenges and applications in the field of civil engineering.

Data Science Introduction: Data and types, Big Data and Distributed Databases, Application and purpose of data, Data Science, The data science process.

Introduction to R and RStudio: Installing and configuring RStudio, R Packages, Basic syntax, variables, Operators, Data types, Control Flow, Sequence Generation (range function), StringOperations, Functions, Loop Functions and Debugging (lapply, apply, mapply, tapply, split, Diagnosing), Simulation & Profiling (Random Number, Linear Model, Random Sampling), File Handling in R (Reading different files in R), Introduction to Swirl, Regular Expression.

Data Cleaning and Summarization: Matrices, Factors, Data Frames, Vectors, Lists, Data Cleaning and reading data from different data source, Reading Large Tables, Subsetting and Sorting, Summarizing Data, Creating New Variables, Reshaping Data, Managing Data Frames with dplyr – Introduction, Managing Data Frames with dplyr - Basic Tools, Merging Data

Data Visualization in R: Setting Your Working Directory (Windows), Principles of Analytic Graphics, Lattice Plotting, Base Plotting System, Plotting using ggplot2/Matplotlib library (Histogram, Box Plot, Scatter Plot, Bar Graphs, Line Graph, etc). Data visualization in civil engineering.

Data Science Advance Topics in R: Basics of Correlation, Regression, Stochastic analysis, Working with Color in R Plots, Dimension Reduction:(Principle Component Analysis, Singular Value Decomposition), Model Evaluation Parameters.

Laboratory Work

Implementation of various data analysis techniques to civil engineering problems

Course Learning Objectives (CLO)

The students will be able to:

1. To manage, manipulate, clean, and analyze different types of data
2. To visualize data using different visualization techniques
3. To develop dashboards for real-time data sets
4. To understand data correlation, reduction, and summarization

Text Books

1. Jiawei Han, Micheline Kamber, Jian Pei, Data Mining Concepts and Techniques, (4th Ed.), Morgan Kaufmann (2022)
2. Roger D. Peng, R Programming for Data Science, Leanpub (2016)

Reference Books

1. Trevor Hastie Robert, Tibshirani Jerome Friedman, The Elements of Statistical Learning, Springer (2017)